Basic Electronics

When we discuss electronics, we are talking about generally small circuits using small amounts of electricity. These circuits handle electricity that is of lesser power by a great margin than that coming out of your wall outlet. Suffice to say, we won't be talking about residential electricity here. Instead we are concerned with small electrical signals used for sending information on the low end, to at the most enough electricity to spin a motor on the high end. To facilitate this conversation, we will start with a basic circuit to build from.

Circuits

The image in Figure 12-1 shows a very simple little circuit.

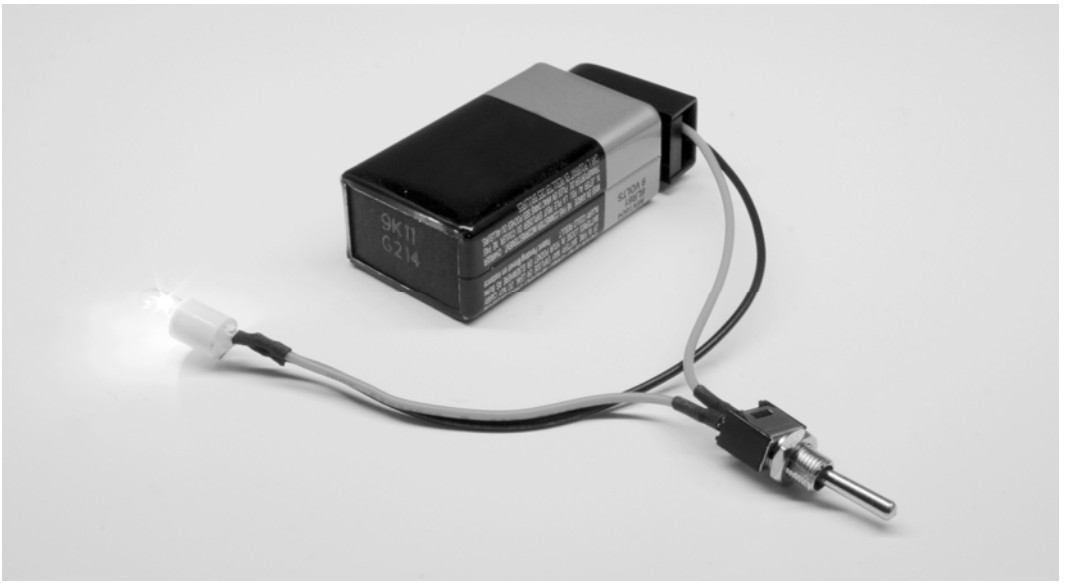


Figure 12-1. *Basic circuit*

This circuit has one job: to light up a lamp when the switch is closed. It has all of the prerequisites for a completed circuit that includes a power source, in this case a 9-volt battery, conductors in the form of insulated wires to carry the electricity, a load that resists the flow of current, here an incandescent lamp that produces light and a little heat in the process, and our circuit has a switch to interrupt the flow of electricity at our whim. A **power source** is needed to provide a source of electricity, whether this comes from the chemical reaction created inside of a battery or the power generated by a power plant down the road. **Conductors**, copper being an especially good one, send the electricity from the circuit's source to its load and back again, connecting each component to make a completed circuit. **Components** like our lamp create a resisting load when placed in the circuit that converts our electricity into something else, whether it's light, heat, movement, or so on. Without a sufficient load on the circuit,